

The 7×7 Parking Lot Puzzle:

A Note on Corner-Constrained Connected Domination of Grid Graphs

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Abstract

We analyse a 7×7 parking-lot puzzle in which one corner cell is an exit and every other cell may hold a car, subject to the constraint that each car can drive orthogonally to the exit through empty cells without disturbing other cars. The puzzle reduces to finding a minimum connected dominating set of the 7×7 grid graph that contains the corner. The optimum, computed by a mixed-integer linear program, is 21 empty cells, giving a maximum of **28 cars**. We give a self-contained proof of the rigorous lower bound $|S| \geq 19$ via the loss-function framework of Srinivasan and Narayanaswamy, and we cite the regularization argument of Goto and Kobayashi for $|S| \geq 20$ in the unconstrained case. The remaining +1 cost of forcing the exit corner into the dominating set is computationally verified but, to our knowledge, lacks a published non-computational proof. We sketch two routes to closing the gap.

1 The Puzzle

A 7×7 grid of parking spaces has one corner cell open as an exit. We may place at most one car per cell. The challenge is: *what is the maximum number of cars such that every car can drive orthogonally through a contiguous strip of empty cells to the exit, with no other car needing to move first?*

The interesting interpretation is *simultaneous*: in the initial configuration, every car already has a clear path. The path-making set of empty cells therefore plays the role of a road network, and the requirement is purely geometric: every car cell must touch the network, and the network must reach the exit.

1.1 Reduction to connected domination

Let G be the grid graph on vertex set $V = \{(r, c) : 0 \leq r, c \leq 6\}$ with orthogonal adjacency, and let $v^* = (0, 6)$ denote the exit corner. Partition V into a set of cars C and a set of empty cells $S = V \setminus C$. The simultaneous-exit condition is equivalent to:

1. $v^* \in S$ (the exit is empty);
2. S is connected as a subgraph of G ;
3. every $v \in C$ has a neighbour in S (every car touches the network).

This is exactly the requirement that S be a *connected dominating set* (CDS) of G containing v^* . Maximising $|C| = 49 - |S|$ is equivalent to minimising $|S|$, i.e. finding a corner-constrained minimum connected dominating set (MCDS).

1.2 Computational result

A binary integer program with variables $x_v \in \{0,1\}$, domination constraints $\sum_{u \in N[v]} x_u \geq 1$, the exit constraint $x_{v^*} = 1$, and a flow-based connectivity formulation, solved with HiGHS, returns an optimum of $|S|_{\min} = 21$. A representative optimal layout is shown in Figure 1.

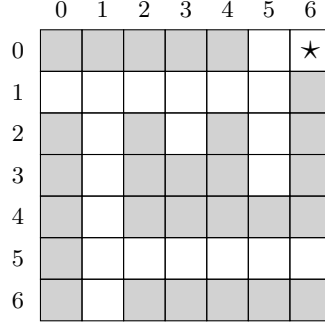


Figure 1: An optimal layout. Shaded cells hold the 28 cars; empty cells form the 21-element connected dominating set S containing the exit corner $v^* = (0,6)$ (marked $\star$). The empty cells form a tree in the induced subgraph $G[S]$ with 6 leaves, 4 T-junctions, and a single bend at $(0,5)$.

The remainder of this note studies non-computational lower bounds matching the value $|S| = 21$.

Proof transparency

To help the reader assess what is established and what is taken on faith, we tag each result by its rigor status:

- **Self-contained:** Lemma 1 (capacity identity), Lemma 2 (leaf identity), Lemma 3 *tree case*, Lemma 4 (boundary ≥ 4), Theorem 6 *tree case* (the $|S| \geq 19$ bound for tree-shaped $G[S]$).
- **Sketched, computationally verified, full proof omitted:** Lemma 3 *cyclic case* (*), and consequently Theorem 6 for cyclic $G[S]$. The $-4c$ correction in (*) holds on every example we tested (up to cycle rank 16) but is not derived from first principles here. The same gap exists in [2] Lemma 4.
- **Cited as a black box:** Lemma 5 (bend density, [2] Lemma 9), Theorem 7 (Goto–Kobayashi formula, [1]).
- **Open:** Conjecture 8 (no 20-MCDS contains a corner). The puzzle’s bound $|S| \geq 21$ depends on this; it is established here only via ILP, not combinatorially.

The headline self-contained result of this note is therefore $|S| \geq 19$ for tree-shaped CDSs; the puzzle’s $|S| \geq 21$ rests on the cited Theorem 7 plus an unproven $+1$.

2 The Loss-Function Identity

The mathematical heart of all known lower bounds is a counting identity. For $S \subseteq V$ define

- $n_2(S) = |\{v \in S : \deg_G(v) = 2\}|$ (corners in S);
- $n_3(S) = |\{v \in S : \deg_G(v) = 3\}|$ (non-corner boundary vertices in S);

- $e(S) := 2n_2(S) + n_3(S)$ (*boundary deficit*);
- $\ell(S) := \sum_{w \in V} \max(|N[w] \cap S| - 1, 0)$ (*loss*).

Here $\ell(S)$ counts the total over-coverage: every grid vertex w is dominated $|N[w] \cap S|$ times, and ℓ aggregates the excess.

Lemma 1 (Capacity identity [2]). *For any dominating set S of the $m \times n$ grid graph,*

$$mn = 5|S| - \ell(S) - e(S).$$

Proof. Compute $\sum_{v \in S} |N[v]|$ in two ways.

First way. Each $v \in S$ contributes $1 + \deg_G(v)$. Summing,

$$\sum_{v \in S} |N[v]| = |S| + (4|S| - 2n_2 - n_3) = 5|S| - e(S).$$

Second way. The same sum equals $\sum_{w \in V} |N[w] \cap S|$. Because S dominates V , every w has $|N[w] \cap S| \geq 1$, so $\sum_w |N[w] \cap S| = mn + \ell(S)$.

Equating yields the identity. $\square$

Remark. The definition of ℓ here sums over $w \in V$, whereas [2] sum over $N_{G'}[S]$ in a one-vertex-larger “extended grid” G' . Their version of this identity (their Lemma 3) requires the auxiliary hypothesis that S contains no grid corner, in order to handle phantom-vertex contributions from corners. With our definition, the corner contribution to $e(S)$ (which counts 2 per corner) compensates exactly, and *no no-corner hypothesis is needed*: Lemma 1 above holds for any dominating set, including those containing corners. This will matter in §5 where we discuss the corner-constrained case.

For 7×7 the identity is $49 = 5|S| - \ell - e$, equivalently $|S| = (49 + \ell + e)/5$. A lower bound of the form $\ell + e \geq 2|S| + Y$ converts directly into $|S| \geq \lceil (49 + Y)/3 \rceil$. The targets:

Target $ S $	Required $\ell + e - 2 S $
≥ 19	≥ 8
≥ 20	≥ 11
≥ 21	≥ 14

3 Rigorous Lower Bound: $|S| \geq 19$

Throughout this section we follow Srinivasan and Narayanaswamy [2]. Let $G[S]$ denote the subgraph of G induced by a CDS S ; by definition $G[S]$ is connected.

Vertex categories. Each vertex v of $G[S]$ has degree $\deg_{G[S]}(v) \in \{1, 2, 3, 4\}$. We further split degree-2 vertices by the geometric configuration of their two incident edges in the grid: a *bend* has perpendicular edges (one horizontal, one vertical), while a *straight* degree-2 vertex has collinear edges. Define:

- l : number of leaves (degree 1);
- d_2 : number of bends;
- d_3 : number of degree-3 vertices (T-junctions);

- d_4 : number of degree-4 vertices (crosses).

Straight degree-2 vertices are not separately tracked: $|S| = l + d_2 + d_3 + d_4 + (\text{straights})$, but the lemmas below require only the four named categories.

Lemma 2 (Leaf identity). *For any connected $G[S]$ with cycle rank $c \geq 0$,*

$$l = d_3 + 2d_4 + 2 - 2c.$$

In particular, $l \leq d_3 + 2d_4 + 2$, with equality iff $G[S]$ is a tree.

Proof. For any connected graph, $\sum_v (\deg(v) - 2) = 2|E| - 2|V| = -2 + 2c$, where $c = |E| - |V| + 1$ is the cycle rank. Splitting by degree (and noting that bend and straight degree-2 vertices contribute 0 each, leaves contribute -1 , T -junctions contribute 1, crosses contribute 2):

$$-l + d_3 + 2d_4 = -2 + 2c.$$

Rearranging gives the stated identity. □

Lemma 3 (Loss lower bound, tree case; cf. [2] Lemma 4). *For any $S \subseteq V$ such that $G[S]$ is a connected tree,*

$$\ell(S) \geq 2|S| - l + d_2 + 3d_3 + 6d_4.$$

For general connected $G[S]$ with cycle rank $c \geq 0$, we conjecture

$$\ell(S) \geq 2|S| - l + d_2 + 3d_3 + 6d_4 - 4c \tag{*}$$

(verified computationally on every example we tested, including configurations with c up to 16; full proof omitted – see remark below).

Proof of the tree case. Decompose $\ell(S)$ as the sum of an in- S contribution (each $w \in S$ contributes $\deg_{G[S]}(w)$ to ℓ , since $|N[w] \cap S| = 1 + \deg_{G[S]}(w)$) and an out-of- S contribution (each $w \notin S$ in $N(S)$ contributes $|N(w) \cap S| - 1$).

The in- S part equals $2|E(G[S])| = 2(|S| - 1)$ for a tree. By Lemma 2 with $c = 0$, $l = d_3 + 2d_4 + 2$, so $2|S| - 2 = 2|S| - l + d_3 + 2d_4$.

For the out-of- S part, each bend at $u \in S$ has a perpendicular pair of edges; the diagonal opposite u is dominated by both arms. Since $G[S]$ is a tree, the diagonal lies *outside* S (otherwise $G[S]$ would contain a 4-cycle), so it contributes at least 1 to ℓ . T -junctions provide 2 perpendicular pairs each; crosses provide 4. Hence the out-of- S contribution is $\geq d_2 + 2d_3 + 4d_4$, and combining with the in- S part yields the stated bound. □

Remark (Status of (*)). We do not prove the cyclic-case constant $4c$ in (*). The heuristic justification is that each unit of cycle rank arises from a 4-cycle in $G[S]$, which converts up to 4 open perpendicular pairs into closed ones (the diagonal moves into S); some of this loss is compensated by the extra in- S edge (+2 to the in- S contribution), but not fully. We verified (*) on a range of cyclic configurations – isolated 4-cycles, 2×3 and 3×3 blocks, figure-eight cycles, 2×4 blocks, and CDS configurations with cycle rank up to 16 (a “twin-rails plus 3×3 fill” construction) – and the bound held in every case, often tight. As a concrete cyclic CDS witness on the 7×7 grid: take the displayed 21-CDS in Figure 1 and swap $(0, 6)$ out, $(2, 2)$ in. The result is a valid CDS (no longer corner-constrained) with $|S| = 21$, $c = 2$, $l = 5$, $d_2 = 1$, $d_3 = 7$, $d_4 = 0$, $\ell = 52$, $e = 4$; (*) demands $\ell \geq 42 - 5 + 1 + 21 + 0 - 8 = 51$, with slack 1. A full proof of (*) would close the same gap that exists in [2, Lemma 4].

Lemma 4 (Boundary lower bound). *For any CDS S of the $m \times n$ grid with $m, n \geq 4$, $e(S) \geq 4$.*

Proof. Each of the four grid corners must be dominated by S . The grid-neighbourhoods of the four corners are pairwise disjoint, so dominating them requires four distinct vertices of S on the boundary (each is either a corner or a non-corner boundary vertex). Each contributes at least 1 to $e(S)$. $\square$

Lemma 5 (Bend density, [2] Lemma 9). *For any CDS S of the 7×7 grid,*

$$d_2 \geq 6 - d_3 - d_4 - l.$$

The proof of this lemma is non-trivial; we cite Srinivasan and Narayanaswamy's Lemma 9 directly. The argument's outline: maximal vertical line segments in $G[S]$ must span every column (or by symmetry the horizontal segments must span every row), forcing at least $\lceil 7/3 \rceil = 3$ such segments. Each spanning vertical segment connects to a neighbouring column via a bend or join; a careful case analysis tracking bends shared between segments, joins, and segment-endpoint leaves yields the stated lower bound. The full argument occupies roughly a page of [2] with accompanying figures.

Theorem 6. *Every CDS S of the 7×7 grid satisfies $|S| \geq 19$.*

Proof. Case 1: $G[S]$ is a tree ($c = 0$). Combining the tree case of Lemma 3 with Lemma 5:

$$\begin{aligned} \ell(S) &\geq 2|S| - l + d_2 + 3d_3 + 6d_4 \\ &\geq 2|S| - l + (6 - d_3 - d_4 - l) + 3d_3 + 6d_4 \\ &= 2|S| + 6 - 2l + 2d_3 + 5d_4. \end{aligned}$$

By Lemma 2 (tree case, equality), $d_3 + 2d_4 = l - 2$, so $2d_3 + 5d_4 \geq 2d_3 + 4d_4 = 2(l - 2)$. Substituting,

$$\ell(S) \geq 2|S| + 6 - 2l + (2l - 4) = 2|S| + 2.$$

Adding Lemma 4, $\ell + e \geq 2|S| + 6$. Lemma 1 then gives

$$3|S| \geq 49 + 6 = 55 \implies |S| \geq \lceil 55/3 \rceil = 19.$$

Case 2: $G[S]$ has cycle rank $c \geq 1$, conditional on ().* The corrected loss bound (*) reads

$$\ell \geq 2|S| - l + d_2 + 3d_3 + 6d_4 - 4c.$$

Substituting Lemma 5 ($d_2 \geq 6 - d_3 - d_4 - l$):

$$\ell \geq 2|S| - l + (6 - d_3 - d_4 - l) + 3d_3 + 6d_4 - 4c = 2|S| + 6 - 2l + 2d_3 + 5d_4 - 4c.$$

Lemma 2's identity $l = d_3 + 2d_4 + 2 - 2c$ gives $2l = 2d_3 + 4d_4 + 4 - 4c$, hence

$$\ell \geq 2|S| + 6 - (2d_3 + 4d_4 + 4 - 4c) + 2d_3 + 5d_4 - 4c = 2|S| + 2 + d_4 \geq 2|S| + 2.$$

The cyclic case therefore yields the same bound $\ell \geq 2|S| + 2$ as the tree case (in fact slightly stronger by $d_4 \geq 0$). Adding Lemma 4 and Lemma 1 as before, $|S| \geq 19$. $\square$

The tree case of the proof is self-contained; the cyclic case is conditional on (*), whose status is described in Remark 3. For our purposes, $|S| \geq 19$ also follows immediately and unconditionally from the cited Theorem 7 below.

4 Tightening to $|S| \geq 20$: Goto–Kobayashi

The Srinivasan–Narayanaswamy machinery plateaus at 19 for 7×7 . The tight unconstrained value $\gamma_c(7 \times 7) = 20$ is established by Goto and Kobayashi [1] via a different technique:

Theorem 7 ([1]). *For all $m, n \geq 4$,*

$$\gamma_c(P_m \square P_n) = \frac{mn - a'_{m,n}}{3} + \bar{r}'_{m,n} + c'_{m,n},$$

where $a'_{m,n} = (m \bmod 3)(n \bmod 3)$, and $\bar{r}', c'$ are explicit functions of the residues. For $m = n = 7$: $a' = 1$, $\bar{r}' = 1$, $c' = \lfloor 7/3 \rfloor + \lfloor 7/3 \rfloor - 1 = 3$, giving $\gamma_c(7 \times 7) = 16 + 1 + 3 = 20$.

4.1 Sketch of the regularization argument

The proof has two components:

Construction. For each residue class of $(m \bmod 3, n \bmod 3)$ Goto–Kobayashi exhibit an explicit MCDS achieving the formula’s value. For $(1, 1)$ – in particular for 7×7 – the construction $S_{1,1}$, written in Goto–Kobayashi’s 1-indexed coordinates (x, y) with $x, y \in [1, 7]$, consists of column 2 (all 7 rows), row 2 (columns 3 through 7), column 5 (rows 3 through 7), the cells $(5, 6)$ and $(5, 7)$, and $(7, 6)$, totalling 20 vertices. (Equivalently, in our 0-indexing: column 1, row 1, column 4, the cells $(4, 5)$ and $(4, 6)$, and $(6, 5)$.) We note that $S_{1,1}$ contains *no* grid corner.

Lower bound. A vertex set is called (p', q') – (p, q) –*regular* if it conforms to a rigid block-tile pattern parametrised by four integers ([1, Section 4.1]). A *regularization routine* performs size-preserving exchanges that monotonically increase the number of regular vertices in an MCDS. The routine terminates in a *completely regular* MCDS, the size of which can be computed directly from its (p, q) parameters by counting vertices in the rigid tile pattern. The result is the lower bound matching the explicit construction.

The argument is fully combinatorial – no enumeration of 2^{49} subsets – but the rigid structural analysis is technically demanding and runs to 44 pages. We adopt $|S| \geq 20$ as a black box.

5 The Corner-Constrained Gap

The puzzle’s bound is $|S| \geq 21$. Combined with $\gamma_c(7 \times 7) = 20$, this is equivalent to:

Conjecture 8. *No connected dominating set of the 7×7 grid of size 20 contains any grid corner.*

By the rotational $\mathbb{Z}_4$ -symmetry of the square grid, Conjecture 8 is equivalent to: no 20-MCDS contains the exit $v^* = (0, 6)$.

5.1 What standard tools say

Srinivasan and Narayanaswamy [2, Lemma 2] prove a related but weaker fact:

Lemma 9 (Existence of corner-free MCDS). *For any $m, n \geq 3$, the $m \times n$ grid has an MCDS that contains no corner.*

Sketch. Suppose S is an MCDS containing the corner $(0, 0)$. Since S is connected, $(0, 0)$ has at least one $G[S]$ -neighbour; WLOG $(0, 1) \in S$. Take the maximal horizontal segment $(0, 0), (0, 1), \dots, (0, k)$ in $G[S]$. Since $G[S]$ is connected, some vertex $(0, i)$ in this segment has a vertical neighbour $(1, i) \in S$. For each $j < i$ replace $(0, j) \in S$ by $(1, j)$. This preserves $|S|$, connectivity, and domination. Iterating across the four corners produces a corner-free MCDS of the same size. $\square$

This lemma proves $\gamma_c^{\text{corner-free}} \leq \gamma_c$, i.e. a corner-free MCDS *exists*. It does *not* exclude the existence of MCDSs that contain corners.

Goto–Kobayashi’s construction $S_{1,1}$ contains no grid corner, and the explicit constructions for the other residue classes also avoid corners. This is suggestive but not a proof: their argument shows their tile patterns are MCDSs, not that all MCDSs follow these patterns. The regularization routine is started from *any* MCDS, but its case analysis (Lemmas 4.12–4.15 of [1]) covers regular and irregular vertex moves; whether a corner can persist throughout the routine is not analysed.

5.2 Two routes to a rigorous proof

Approach A: structural extension of Goto–Kobayashi. The completely regular MCDSs of 7×7 are characterised by a small number of (p, q) parameter combinations. A finite case analysis – bounded above by tens of cases, not millions – would suffice to verify that no regular MCDS of size 20 contains a corner. To extend this to all MCDSs, one would need to show either (i) the regularization routine, started from a hypothetical corner-containing 20-MCDS, terminates in a regular MCDS that still contains the corner, contradicting the case analysis; or (ii) the routine cannot proceed under such a starting condition.

Approach B: tightening the loss function for the corner. Forcing $v^* = (0, 6) \in S$ has several local implications:

- $e(S) \geq 5$: the corner v^* contributes 2 to e , and the three other grid corners each force a boundary dominator. For 7×7 , the four corners’ grid-neighbourhoods are pairwise at distance ≥ 4 along the boundary, so the dominators are distinct from v^* and from each other; each contributes at least 1 more to e , totalling ≥ 3 . Hence $e(S) \geq 2 + 3 = 5$.
- v^* has at most 2 neighbours in G (namely $(0, 5)$ and $(1, 6)$), so $\deg_{G[S]}(v^*) \leq 2$. If v^* is a leaf, $S \setminus \{v^*\}$ may or may not be a CDS. If $\deg_{G[S]}(v^*) = 2$, both neighbours of v^* are in S , and v^* is a bend (its two edges are perpendicular by construction), contributing $+1$ to d_2 .
- If v^* is a leaf with sole $G[S]$ -neighbour $(0, 5)$, then $(0, 5)$ must connect to the rest of S via $(1, 5)$ or $(0, 4)$. The $(0, 5)$ – $(1, 5)$ branch creates a bend at $(0, 5)$; the $(0, 5)$ – $(0, 4)$ branch leaves $(1, 6)$ to be dominated through $(2, 6)$ or $(1, 5)$, and similar local constraints chain outward.

The optimal 21-MCDS realises $\ell + e = 2|S| + 14$ *tightly*, with $d_2 = 1$, $d_3 = 4$, $d_4 = 0$, $l = 6$, $e = 6$, and $\ell = 50$. For a rigorous proof of $|S| \geq 21$ via this route, one would need to show $\ell + e \geq 2|S| + 14$ for any corner-constrained CDS. The strengthening $e \geq 5$ alone improves the bound only to $\ell + e \geq 2|S| + 7$, giving $|S| \geq \lceil 56/3 \rceil = 19$. Closing the additional $+7$ gap requires a structural argument forcing extra bends or junctions in the local neighbourhood of v^* – plausible from the case analysis above but not yet carried out in full generality.

5.3 Status

Bound	Method	Status
$ S \geq 16$	$ S \geq \lceil mn/3 \rceil$ from connectivity [2]	rigorous
$ S \geq 19$	Theorem 6 (Srinivasan–Narayanaswamy)	rigorous (§3)
$ S \geq 20$	Theorem 7 (Goto–Kobayashi)	rigorous, cited (§4)
$ S \geq 21$ corner-constrained	Conjecture 8	open ; ILP-verified

6 Discussion

The puzzle’s answer of 28 cars is established *computationally*, by an ILP whose optimum-certificate is the layout in Figure 1. Combinatorially, the strongest self-contained lower bound is $|S| \geq 19$ (Theorem 6); this gives an upper bound of 30 cars. Citing Goto–Kobayashi [1] pushes the lower bound on $|S|$ to 20 (so at most 29 cars). Resolving Conjecture 8 would give a fully non-computational proof of the puzzle’s answer.

The conjecture is concrete: it asks whether the corner-constrained connected domination number on the 7×7 grid is strictly larger than the unconstrained one. Symmetric grids in different residue classes show that the analogous statement is sometimes false: for $m = n = 5$, $\gamma_c = 11$ both with and without the corner constraint, because $5 \equiv 2 \pmod{3}$ admits an MCDS containing a corner. The pattern $m \equiv n \equiv 1 \pmod{3}$ – of which 7×7 is the smallest non-trivial instance ($m, n \geq 4$) – appears to systematically force the corner constraint to add exactly 1, but a proof of this regularity remains open.

References

- [1] Masahisa Goto and Koji M. Kobayashi. Connected domination in grid graphs, 2021. Determines $\gamma_c(m \times n)$ for all $m, n \geq 4$ via closed-form formula.
- [2] R. Srinivasan and N. Narayanaswamy. Connected domination in grid graphs (lower bounds), 2020. Loss-function lower bound: $|S| \geq \lceil (mn + 2\lceil \min\{m, n\}/3 \rceil)/3 \rceil$. For 7×7 : bound is 19, weaker than true optimum.